

Two simple exercises to understand change of basis formulas

1) Conversion of a quantity from one unit to another

	Old unit	New unit
1a) Distance	$B = \text{meter}$	$B' = \text{kilometer}$
1b) Time	$C = \text{second}$	$C' = \text{hour}$

You want to change from old unit to new unit. How to convert?

You need a conversion factor. From which unit to which?

Let's adopt the view that it makes more sense to express the new unit in terms of the old one. Do this for both 1a and 1b:

Conversion factor for 1a) distance is $Q =$
Conversion factor for 1b) time is $P =$

- **Write** the equations carefully, including units! Don't just eyeball it.
- How to convert a measurement from old units to new units?
Write an equation: new measure = ...

2) Conversion of a rate

Let's say the speed is $A = 5 \text{ meter/second}$. How to convert this speed to new units km/hour? Of course you know how to do this. But

- **Write** an answer in terms of GIVEN LETTERS.

The formula will be a product of three things, which should be written in the exact same order in which you use them during the conversion process, but written from right to left.

How this is relevant for us

This is the one-dimensional case of change of basis.

1) amounts to changing basis in one vector space from B to B' (respectively C to C')

2) Distance = speed $\times$ time is like $y = Ax$, where speed is the matrix of the linear map connecting the two quantities.

The task here is to find the matrix describing the same relationship when we change units on either side.